

Definition 1. The **least common multiple** of non-zero integers a and b , denoted $\text{lcm}(a, b)$, is the smallest integer in $\mathbb{N}$ that is multiple of both a and b .

Proposition 1. If a, b, c are positive integers, then $\text{lcm}(ac, bc) = c \cdot \text{lcm}(a, b)$.

Solution. Let a, b, c be positive integers, and let $m = \text{lcm}(ac, bc)$ and $n = c \cdot \text{lcm}(a, b)$.

On the one hand, $\text{lcm}(a, b) = ax = by$ for some $x, y \in \mathbb{Z}$. So $n = acx = bcy$ is a multiple of both ac and bc . Since m is the smallest such number, $m \leq n$.

On the other hand, $m = acx = bcy$ for some $x, y \in \mathbb{Z}$. So $\frac{m}{c} = ax = by$ is a multiple of both a, b . Since $\frac{n}{c}$ is the smallest such number, $\frac{n}{c} \leq \frac{m}{c}$, i.e., $n \leq m$.

Therefore, we've shown that $n \leq m$ and $m \leq n$, hence $n = m$.

Definition 2. The **greatest common divisor** of integers a and b , denoted $\text{gcd}(a, b)$, is the largest integer that divides both a and b .

Definition 3. Two integers a and b are said to be relatively prime if $\text{gcd}(a, b) = 1$. (or coprimes)

Proposition 2. Let p, q be prime numbers. If $\text{gcd}(p, q) > 1$, then $p = q$.

Solution. Suppose $n = \text{gcd}(p, q) > 1$. Note that $n \mid p$, so $n = 1$ or $n = p$. Likewise, $n = 1$ or $n = q$. Since $n > 1$, it follows that $n = p$ and $n = q$. Thus, $p = q$.

Theorem 1 (Bezout's Identity). For any $a, b \in \mathbb{Z}$, there exists $x, y \in \mathbb{Z}$ such that

$$\text{gcd}(a, b) = ax + by.$$

Solution. Let $a, b \in \mathbb{Z}$, and define $d = \min \{ax + by : x, y \in \mathbb{Z}, ax + by > 0\}$ by the Well-ordering Principle.

d is a divisor: By the division algorithm, $a = qd + r$ with $0 \leq r < d$. Note that

$$r = a - qd = a - q(ax + by) = a(1 - qx) + b(-qy)$$

so $r = aX + bY$. Since d is the smallest positive, $r = 0$. So $d \mid a$ since $a = qd$. Homework: change the letter to prove that $d \mid b$.

d is the greatest: Suppose that d' is another divisor. Our goal is to show that $d' \leq d$. Since $d' \mid a$ and $d' \mid b$, it follows that $a = d'm$ and $b = d'n$. So

$$d = ax + by = (d'm)x + (d'n)y = d'(mx + ny),$$

do $d' = d/(mx + ny) \leq d$

Counterexamples

The idea is to prove that the negation of “If P, then Q” is true, i.e., “P and not Q”. Analogously, “for all x such that P, then Q” becomes “there exists x such that P and not Q”.

Example 1. For every $n \in \mathbb{N}$, the integer $2^n - 1$ is prime. $n = 11 \implies 2^n - 1 = 2047 = 23 \cdot 89$

Example 2. If A, B and C are sets, then $A - (B \cap C) = (A - B) \cap (A - C)$. $A = \{1, 2, 3\}$, $B = \{1, 2\}$, $C = \{2, 3\}$.

Example 3. If A, B and C are sets, then $A - (B \cup C) = (A - B) \cup (A - C)$. $A = \{1, 2, 3\}$, $B = \{1, 2\}$, $C = \{2, 3\}$.

Example 4. If $x, y \in \mathbb{R}$, then $|x + y| = |x| + |y|$. $x = 1$, $y = -1$

Example 5. If $n \in \mathbb{Z}$ and $n^5 - n$ is even, then n is even. $n = 1$

Example 6. If $a, b \in \mathbb{Z}$, $a \mid b$ and $b \mid a$, then $a = b$. $a = 1$, $b = -1$

Example 7. For any irrational numbers a and b , the number a^b is irrational. $a = \sqrt{2}^{\sqrt{2}}$, $b = \sqrt{2}$

Example 8. All functions $f : \mathbb{R} \rightarrow \mathbb{R}$ are continuous.

$$f(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases}$$

Workshop/group activity